

ChristosHadjinikolis

Engineering Manager | AI Harnesses & Production ML Platforms

Ph.D. Computer Science | UCL Research Associate | AI Standards Expert @ ISO/CEN-CENELEC JTC'21

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mediums



leadership

team leadership
coaching
hiring loops
progression
cross-functional delivery

AI harness

Promet (private)
tool-using assistants
prompt contracts
approvals/safety
trace/replay

platform

production ML
internal platforms
streaming systems
service reliability
roadmap execution

trust

replay/evaluation
observability
model/data versioning
auditability
safe AI adoption

operations

runbooks/rollback
shared on-call
MTTR under 30 min
DORA/Jira signals

technical

Java/Kotlin
Python/SQL
Kafka/Flink
MLflow/FastAPI
AWS/GCP

tooling

AI-assisted review
PyPI: dynamicio
skeleton-replay
JetBrains plugin

about me

Engineering Manager for production ML, streaming, and AI platform systems with 16 years across backend engineering, data infrastructure, consulting, AI research, and teaching. At Vortexa, own a major **client-facing ML and streaming estate**, lead a **10-person cross-functional team**, and drove Data Production Team growth from **4 to 30+ people** through hiring loops, system-design culture, platform choices, and delivery discipline. Current private work on **Promet**, a local-first AI harness shell for memory-aware, voice-enabled, tool-using assistants, keeps me hands-on with orchestration, local runtimes, prompt contracts, approvals, traceability, and replay. I turn research-heavy AI/data work into reliable platforms and operating models.

work-experience

Vortexa Ltd, London, UK 12/2020–present

Engineering Manager | ML Systems Lead

Own the **engineering strategy and delivery** for Vortexa's **live ML and streaming intelligence estate**, a major client-facing backend area powering product and commercial workflows, filtering roughly **6M vessel-position records/hour** into production intelligence for **13.5K monitored vessels**. Lead delivery and live-processing strategy with a **10-person cross-functional team** spanning **6 MLE/DE/DS, 2 Product, and 2 SMEs**. The estate acts as an internal platform: downstream teams and product surfaces depend on its correctness, freshness, reliability, and operational clarity.

- **People:** Directly manage the MLE/DE/DS core and lead a **10-person cross-functional delivery team**; hiring manager for **6 roles**, interviewing **31+ candidates** in that capacity and **60+ overall**; retained the full current team, with every member recently promoted.
- **Organisation Scaling:** Drove Data Production Team growth from **4 to 30+** through hiring loops, system-design evaluations, onboarding discipline, mentoring, reduced key-person risk, and clearer ownership boundaries.
- **Service Ownership:** Protect customer-facing reliability with observability, monitoring, runbooks, rollback/fallback plans, Jira alert workflows, and shared on-call ownership while keeping **MTTR under 30 min**.
- **Delivery & Stakeholders:** Partner with Product, analysts, SMEs, and engineering stakeholders to turn product demand, ambiguous semantics, and quality concerns into measurable workstreams and platform priorities.
- **Architectural Inflection Point:** Led the two-year **Kafka Streams-to-Flink migration** for critical live pipelines, reducing partition-coupled scaling and maintenance effort through Flink's dataflow model, independent state/checkpointing, operational UI, and framework-driven design.
- **Production ML Platform:** Established Vortexa's ML lifecycle from research to production: prototyping discipline, **MLflow**, model/data versioning, service-based deployment, replay/evaluation interfaces, live accuracy-monitoring dashboards, and explainability/auditability practices.
- **Engineering Operating Model:** Scale delivery through reusable Flink/Airflow patterns, Backstage adoption, dev containers, ADRs, local E2E/integration tests, DORA/Jira signals, and a weekly architecture forum; cut a **3h DAG feedback loop to under 5 min**.

DataReply, London, UK 04/2016–12/2020

Senior Consultant | Data Scientist / ML Engineer

Joined Data Reply's London spin-off as its first consultant and played a formative role in growing it from **three practitioners and two managers to 30+ consultants**. Progressed to Senior Consultant; led client meetings, scoped goals, placed consultants, interviewed/mentored hires, and delivered data-platform and ML engineering work across UBS, Vodafone, CNHi and other enterprise clients. Ran the London Flink Meetup for five years.

- **Client Delivery Leadership:** Drove agendas with client stakeholders and product teams, agreed goals, facilitated technical delivery, and supported hiring/consultant placement across banking, telecoms, manufacturing, publishing, and gaming accounts.
- **Vodafone Group:** Led technical delivery across multiple cloud/data workstreams; worked on Infinity, a GCP/**Kubeflow** DS platform, shaping CI/CD, workflow automation, ML lifecycle support, and feature-engineering migration practices.
- **CNHi:** Lead Data Scientist/Scrum Master on predictive maintenance; managed/guided a DS team using **PySpark** over live vehicle sensory data with product-facing backlog discipline.
- **UBS:** Grew from Data Scientist into ML Engineer through XP/pairing while delivering graph/process-mining analytics, observability work, and real-time pipelines with **Kafka**, **Elasticsearch**, and Python.

Roles in UK Institutions 2010–2016

Coding Teacher @DGC | Teaching Assistant @KCL/UCL

Built the communication and presentation discipline I now rely on as a leader by teaching Java/Python and core computer science at David Game College from 09/2012 to 04/2016 while completing doctoral research; Teaching Assistant at KCL and UCL.

selected platform/tooling work

Private	Promet Local-first AI workbench/harness shell for memory-aware, voice-enabled, tool-using assistants; explores local run-times, prompt contracts, MCP-style connectors, approval/safety boundaries, traceability, and replay.	Local AI harness
PyPI	skeleton-replay / IDE plugin Python tool and free JetBrains plugin that turn script/pytest runs into traces, architecture snapshots, workflow evidence, and replayable reports for review, onboarding, debugging, and LLM-assisted understanding.	Runtime evidence
PyPI	dynamicio Free public tooling for making I/O seams explicit, validating schema/data expectations, and switching local/dev/prod datasets so pipeline logic can be tested with faster feedback.	Data contracts

committees/affiliations

since 10/2024	University College London, Dept. of Information Studies Research/teach practical AI standardisation: AI Act, auditability, model/data versioning, explainability, and safe adoption.	Associate Researcher
since 01/2021	ISO - International Organisation for Standardisation Member of JTC'21 Working Group 3, developing AI standards aligned with EU policy and international norms.	Committee Expert Member for JTC'21

education

2010–2014	Ph.D. in Computer Science <i>Thesis: Persuasion Dialogues & Opponent Modelling.</i> Developed logical inference methods for large Knowledge Graphs , combining symbolic reasoning, graph analytics , and Bayesian techniques (e.g., MCMC). Research contributed to scalable AI systems grounded in formal semantics. Received Best Poster Award at IJCAI 2013 (413 submissions). Awarded EPSRC PhD Scholarship , Outstanding TA Award (student-nominated), and earned the Graduate Certificate in Academic Practice . Access Publications on Google Scholar	King's College London, UK
2004–2010	Diploma (BEng) in Computer Engineering Five-year polytechnic degree in Computer Science and Engineering with a strong focus on mathematics (10 courses), covering theoretical aspects, networking, and telecommunications. Majored in Artificial Intelligence .	University of Thessaly, GR
2002–2004	Military Officer: Second Lieutenant Completed national service as a commissioned Second Lieutenant.	National Guard, CY

talks/interviews

2023	Bill Raymonds, Agile in Action Agile Data Science: Sailing Through Data Science: The Agile Journey at Vortexa, Details: YouTube	Podcast Interview
2022	Open Data Science Conference dynamicio , a published PyPI library for abstracting I/O in ML/data workflows, Details: here	Industry Conference Talk
2020	iunera: Big Data Innovators The Agile approach in data science explained by an ML expert, Details: here	Interview Blog Post
2020	Big Data Warsaw Monitoring & Analysing Communication and Trade Events as Graphs, Details: here	Industry Conference Talk
2018	Connected Data London AI tribes and interconnections: what's graph got to do with it?, Details: here	Panelist @ Industry Conference
2018	Minds Mastering Machines A talk on doing Data Science the Agile Way, Details: here	Industry Conference Talk

certificates & awards

07/2020	AWS Certified Machine Learning - Specialty Validation Number: D0R0WC7CN1VQ13SV	AWS
01/2019	Process Mining License number: AFU2VTP6273V	Coursera Course Certificates
10/2017	Google Certified Professional - Data Engineer Credential ID: 000000400	Google Cloud Certified
07/2017	Advanced Elasticsearch: Data modelling Enrolment Id: 33442	Elastic
06/2016	Graph Analytics for Big Data License number: NZADYAQ7N8FW	Coursera Course Certificates
05/2016	Apache Spark Developer Certification License number: 1.x-0548	O'Reilly Media
04/2016	Neo4j Certified Professional License number: here	Neo Technology, Creators of Neo4j